

Olist MYSQL Analysis Capstone

1. Data Overview

The Olist dataset captures real e-commerce transactions from 2016 to 2018, covering multiple tables such as orders, payments, customers, products, and sellers. Each table links through unique IDs, allowing full visibility of customer behavior and revenue flow. The data is clean and consistent, with timestamps enabling monthly and yearly trend analysis. Minor gaps in category naming exist but do not affect overall insights.

2. Three Business Insights

Insight 1 – Revenue Growth and Seasonality (Q2–Q3) Monthly revenue steadily increased from early 2017 through mid-2018, with sharp peaks in November and mid-year months. These surges align with holiday and promotional periods, showing strong seasonal buying behavior. The month-over-month analysis confirms predictable cycles of growth and slowdown, useful for forecasting and campaign timing.

Insight 2 – Category Concentration (Q4–Q8) A few product categories dominate total revenue, with the top category contributing roughly 10–15% of all sales. This concentration indicates that Olist's business performance relies heavily on a limited set of high-demand categories. Diversification could reduce risk and stabilize revenue across seasons.

Insight 3 – Customer Value and Retention (Q1, Q6, Q7) High-spending customers represent a small portion of the base but drive most of the revenue. Most buyers purchase only once, while repeat customers contribute disproportionately to total sales. Segmenting customers into low, medium, and high spend tiers enables targeted retention and loyalty strategies.

3. Two Recommendations

Recommendation 1 – Strengthen Retention Programs Focus marketing efforts on medium and high-spend customers through loyalty rewards, personalized offers, and post-purchase engagement. Increasing repeat purchase rates will yield higher lifetime value and stabilize monthly revenue.

Recommendation 2 – Diversify Product Portfolio Reduce dependency on top categories by promoting emerging ones and onboarding new sellers. This will balance revenue streams and protect the business from seasonal or category-specific downturns.

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